

Deep Learning Reveals Endogenous Sterols as Allosteric Modulators of the GPCR-G α Interface

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eLife Assessment

The authors present a computational pipeline for the identification of endogenous allosteric modulators of GPCRs, with experimental validation performed in a yeast system. This approach is **valuable** for a broad audience, including GPCR structural biologists, molecular pharmacologists, and computational biophysicists. However, the rigor of the computational methods needs to be strengthened to provide stronger evidence for the study's conclusions, which is currently **incomplete**. The authors should justify their methodological choices and provide greater detail and clarity regarding each computational layer of the pipeline.

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Abstract

Endogenous intracellular allosteric modulators of GPCRs remain largely unexplored, with limited binding and phenotype data available. This gap arises from the lack of robust computational methods for unbiased cavity identification, cavity-specific ligand design, synthesis, and validation across GPCR topology. Here, we developed Gcoupler, an AI-driven generalized computational toolkit that leverages an integrative approach combining *de novo* ligand design, statistical methods, Graph Neural Networks, and Bioactivity-based ligand prioritization for rationally predicting high-affinity ligands. Using Gcoupler, we interrogated intracellular metabolites that target and regulate the GPCR-G α interface (Ste2p-Gpa1p), affecting pheromone-induced programmed cell death in yeast. Our computational analysis, complemented by experimental validations, including genetic screening, multi-omics, site-directed mutagenesis, biochemical assays, and physiological readouts, identified endogenous hydrophobic metabolites, notably sterols, as direct intracellular allosteric modulators of Ste2p. Molecular simulations coupled with biochemical signaling assessment in site-directed Ste2p mutants further confirmed metabolites binding to GPCR-G α obstruct downstream

signaling, possibly via cohesive effect. Finally, by utilizing isoproterenol-induced, GPCR-mediated human and neonatal rat cardiac hypertrophy models, we observed elevated metabolite levels attenuate hypertrophic response, reinforcing the evolutionary relevance of this mechanism.

Introduction

G protein-coupled receptors (GPCRs) are critical regulators of cellular processes and thus represent prime drug targets (1–3). While traditional GPCR-targeted therapies focus on orthosteric sites, recent advances have revealed allosteric sites offering novel therapeutic avenues (4, 5)(5–7). Although exogenous synthetic allosteric modulators are known, endogenous counterparts remain poorly characterized (8–13). Developing high-affinity endogenous modulators requires integrating structure-based design, artificial intelligence (AI), and assays, yet traditional approaches like SAR analysis are hampered by limited GPCR allosteric modulator data (14, 15). While experimental techniques like FRET and BRET can validate allosteric compounds, their use in high-throughput screening for novel intracellular modulators is challenging (16–18). Identifying endogenous GPCR allosteric modulators is further complicated by factors like incomplete GPCR topology data (19) and vast chemical space (1, 20). This necessitates a hybrid computational approach combining allosteric site prediction, *de novo* ligand synthesis, and efficient screening, potentially enhanced by AI. Existing *de novo* drug design tools often lack practical applicability for this purpose due to computational limitations and high technical demands (21–27).

To overcome these challenges, we developed Gcoupler, a software suite (available as a Python package and Docker image) that integrates structural biology, statistical methods, and deep learning to identify GPCR allosteric modulators. We demonstrated the usability and applicability of Gcoupler in identifying novel endogenous modulators of GPCRs by exploiting the α -pheromone (α -factor)-induced mating or programmed cell death (PCD) pathway of *S. cerevisiae*. Notably, it is well documented that the highly elevated, non-physiological levels of α -factor trigger PCD in less than half of the MATa population (28). Moreover, an equivalent concentration of α -factor triggers distinct PCD kinetics across distinct laboratory strains; for example, BY4741 is more resistant than the W303 strain (29). We, therefore, hypothesized that a subset of pheromone-resistant cells might regulate the Ste2p-mediated PCD signaling via the endogenous intracellular metabolites by operating at the Ste2-G α binding interface. Using Gcoupler, we identified a subset of intracellular metabolites that could potentially bind to Ste2p (GPCR) at the Gpa1 (G α) binding interface and obstruct the downstream signaling. Our computational results further suggest that hydrophobic ligands such as sterols strengthen the Ste2p-Gpa1p binding and might trigger a cohesive response that potentially obstructs downstream signaling. Experimental evidence further supported these findings that the elevated intracellular levels of these metabolites rescue the pheromone-induced PCD. To evaluate the evolutionary conservation and possible clinically relevant translation of this mechanism, we tested these metabolites on human and rat isoproterenol-induced, GPCR-mediated cardiac hypertrophy model systems and observed attenuated response in the cardiomyocytes pretreated with GPCR-G α -protein interface modulating metabolites.

Results

Development and validation of Gcoupler

Designing novel target molecules by integrating the topological, chemical, and physical attributes of protein cavities necessitates advanced neural networks. While existing approaches like Bicyclic Generative Adversarial Networks (BicycleGANs) (30) and Recurrent Neural Network (RNNs)

(31 [↗](#)) have demonstrated potential, end-to-end standalone tools for GPCR-specific ligand design remain scarce. To address this, we developed Gcoupler, a Python package and Docker image for structure-based, cavity-dependent *de novo* ligand design, employing advanced Graph Neural Networks (GNNs) and robust statistical methods. Gcoupler consists of four interconnected modules, i.e., Synthesizer, Authenticator, Generator, and BioRanker, all together offering an intuitive workflow for ligand design, screening, and prioritization.

The Synthesizer module identifies putative cavities in protein structures, providing users with flexibility to select cavities based on druggability scores or user-supplied critical residues. Leveraging LigBuilder V3 (32 [↗](#)), it generates cavity-specific ligands influenced by topology and pharmacophores, outputting SMILES, cavity coordinates, and other requisite files to downstream modules for further steps (**Figure 1a** [↗](#)). By default, it synthesizes 500 molecules using a hybrid Growing-Linking mode optimized through genetic algorithms though it can also be user-defined. Notably, the fragment library of LigBuilder, comprising of 177 distinct molecular fragments in Mol2 format, allows the selection of multiple seed structures and extensions that best complement the cavity pharmacophores throughout multiple iterative runs. The Synthesizer module of Gcoupler enhances LigBuilder V3 for GPCR ligand design but lacks user-defined library screening, and proposes synthetically challenging molecules, and often requires post-processing to isolate high-affinity binders from a broad affinity range of synthetically designed compounds.

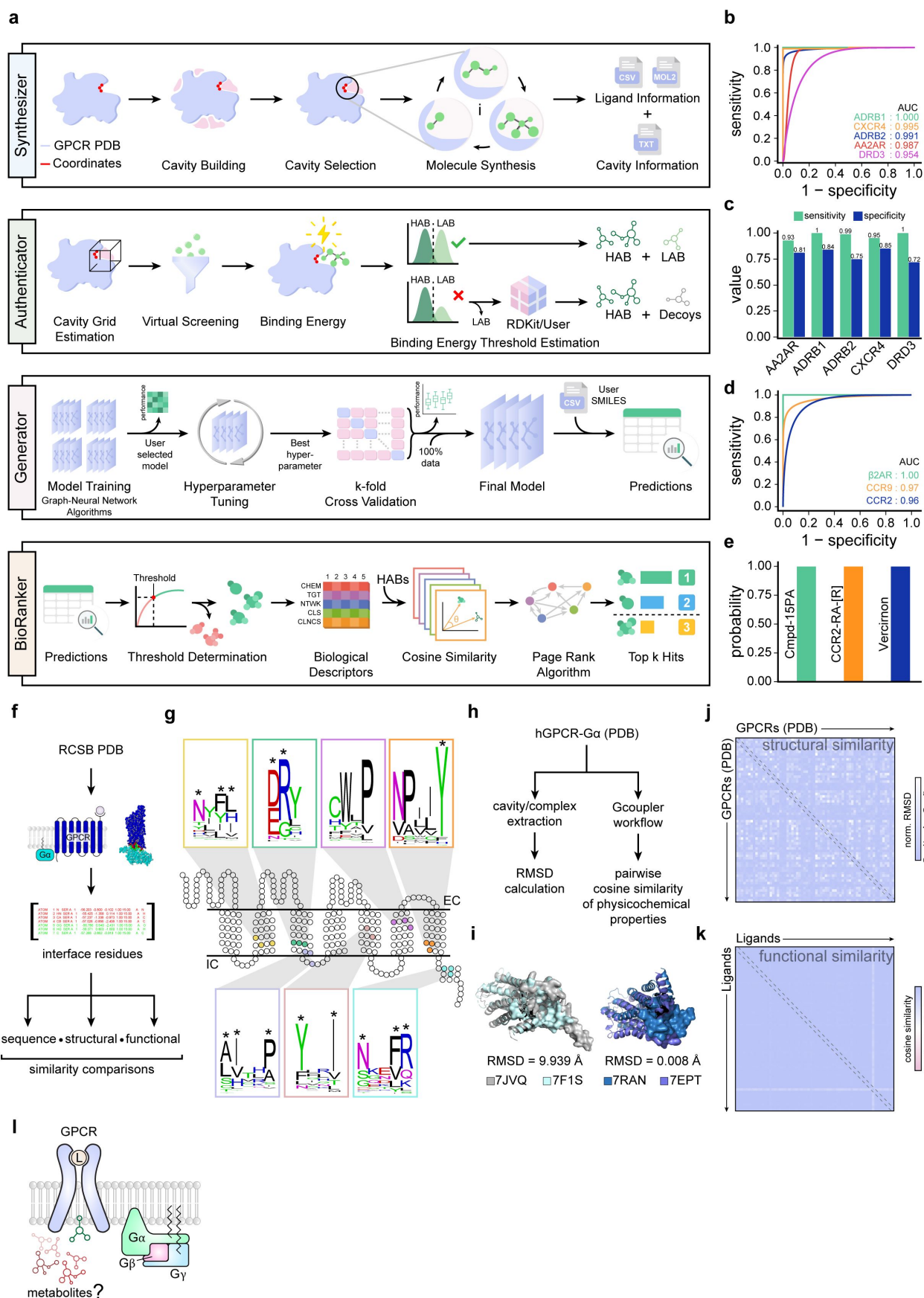


Figure 1

Development, Benchmarking and Validation of Gcoupler Computational Framework.

(a) Schematic workflow depicting different modules of the Gcoupler package. Of note, Gcoupler possesses four major modules, i.e., Synthesizer, Authenticator, Generator and BioRanker. **(b)** AUC-ROC curves of the finally selected model for each of the indicated GPCRs. Note: Experimentally validated active ligands and decoys were used in the testing dataset. **(c)** Bar graphs depicting the sensitivities and specificities of the indicated GPCRs with experimentally validated active ligands and reported decoys. **(d)** The AUC-ROC curve indicating the model performance in the indicated conditions. **(e)** Bar graphs indicating the prediction probabilities for each indicated experimentally validated ligand. **(f)** Schematic workflow illustrates the steps in measuring and comparing the structural conservation of the GPCR-G α -protein interfaces across human GPCRs. **(g)** Snakeplot depicting the standard human GPCR two-dimensional sequence level information. Conserved motifs of the GPCR-G α -protein interfaces are depicted as WebLogo. Asterisks represent residues of conserved motifs present in the GPCRs-G α -protein interfaces. Of note, the location of the motifs indicated in the exemplary GPCR snake plot is approximated. **(h)** Schematic workflow illustrates the steps in measuring and comparing the structural conservation of the GPCR-G α -protein interfaces across human GPCRs. **(i)** Representative structures of the proteins depicting highly conserved (low RMSD) and highly divergent (high RMSD) GPCR-G α -protein interfaces. PDB accession numbers are indicated at the bottom. **(j)** Heatmap depicting the RMSD values obtained by comparing all the GPCR-G α -protein interfaces of the available human GPCRs from the protein databank. Of note, the RMSD of the G α -protein cavity was normalized with the RMSDs of the respective whole proteins across all pairwise comparisons. **(k)** Heatmap depicting the pairwise cosine similarities between the *in silico* synthesized ligands of the GPCR-G α -protein interfaces of the available human GPCRs using Gcoupler. **(l)** Schematic diagram depicting the hypothesis that the intracellular metabolites could allosterically modulate the GPCR-G α interaction.

To address these challenges, we developed Authenticator which validates Synthesizer outputs by segregating high-and low-affinity binders (HABs and LABs) *via* virtual screening using the AutoDock Vina (33) and statistically testing the differences in the binding energies distributions using the Kolmogorov–Smirnov, Epps-Singleton, and Anderson-Darling Test. This module refines synthetic ligand pools by analyzing binding energy distributions, facilitating informed segregation while allowing user-defined thresholds and hypothesis testing for flexibility. The Generator, the third module, employs state-of-the-art GNN models such as Graph Convolution Model (GCM), Graph Convolution Network (GCN), Attentive FP (AFP), and Graph Attention Network (GAT) to construct predictive classifiers using Authenticator-informed classes. With automated hyperparameter tuning and k-fold cross-validation, it ensures robust model training and minimizes overfitting. Notably, by default, Gcoupler employs three-fold cross-validation, but users can adjust this parameter.

Finally, BioRanker, the last module prioritizes ligands through statistical and bioactivity-based tools. The first level ranking offered by BioRanker is composed of a statistical tool that encompasses two distinct algorithms, namely G-means and Youden's J statistics, to assist users in identifying the optimal probability threshold, thereby refining the selection of high-confidence hit compounds (Supplementary Figure 1a). Additionally, bioactivity embeddings computed via Signaturizer (34) enable multi-activity-based ranking using a modified PageRank algorithm. Taken together, Gcoupler is a versatile platform supporting user-defined inputs, third-party tools for cavity selection, and customizable statistical analyses, enhancing its adaptability for diverse ligand design and screening tasks. This integrated framework streamlines cavity-specific ligand design, screening, and ranking, providing a comprehensive solution for GPCR-targeted drug discovery.

To evaluate Gcoupler's performance, we tested its modules across five GPCRs (AA2AR, ADRB1, ADRB2, CXCR4, and DRD3) using experimentally validated ligands and matched decoys from the DUD-E dataset (35). Gcoupler accurately identified orthosteric ligand-binding sites and additional allosteric cavities across all targets, validating its *de novo* cavity detection algorithm (Supplementary Figure 1b). For orthosteric sites, the Synthesizer module generated ~500 compounds per GPCR, and the Authenticator module classified these into high- and low-affinity binders using thresholds optimized for each GPCR (-7 kcal/mol for AA2AR, CXCR4, DRD3; -8 kcal/mol for ADRB1, ADRB2) (Supplementary Figure 1c). Statistical validation confirmed significant separation between these groups ($p < 0.0001$), enabling the Generator module to construct graph-based classification models with high values of AUC-ROC (>0.95), sensitivity, and specificity (Figure 1b-c, Supplementary Figure d-e). These models reliably distinguished ligands from decoys, demonstrating Gcoupler's accuracy in identifying high-affinity ligands. We extended this validation to allosteric sites using three GPCR-ligand complexes (β 2AR-Cmpd-15PA, CCR2-CCR2-RA-[R], CCR9-Vercirnon) from the PDB (36). Gcoupler successfully identified allosteric binding sites and generated classification models for synthetic compounds with consistently high AUC-ROC values (>0.95) (Figure 1d, Supplementary Figure 2a-c). Projection of experimentally validated ligands onto these models further confirmed their predictive accuracy (Figure 1e), underscoring Gcoupler's robustness and versatility for orthosteric and allosteric ligand discovery. We next benchmarked Gcoupler's efficiency against AutoDock using the alpha-1A adrenergic receptor (993 agonists from ChEMBL31). Gcoupler was 13.5 times faster, leveraging its deep learning-based Generator module and AutoDock Vina's efficiency. Both methods provided comparable predictions for active compounds, demonstrating Gcoupler's speed and accuracy, making it ideal for large-scale ligand design and drug discovery (Supplementary Figure 3a-h).

Finally we used Gcoupler to evaluate the ligand space conservation (functional conservation) of the GPCR-Ga interface. We analyzed multiple human GPCR-Ga complexes from the PDB (Figure 1f, Supplementary Table 1), identified conserved motifs (DRY, CWxL, NPxxY) and binding pockets through sequence and structural analyses (Figure 1g). Structural conservation was quantified by normalized RMSD values (mean: 1.47 Å; median: 0.86 Å), confirmed robust topological similarity at the GPCR-Ga interfaces (Figure 1h-j, Supplementary Figure 3i-k, Supplementary Table 2). Using Gcoupler, ~50 ligands per GPCR were synthesized, with physicochemical property analysis (via Mordred) and revealed high cosine similarity, which further supports the functional conservation of GPCR-Ga interface (Figure 1h, k, Supplementary Figure 3l, Supplementary Table 3). In summary, we used Gcoupler to systematically evaluate and analyze the ligand profiles of the GPCR-Ga-protein interface and observed a higher degree of sequence, topological, and functional conservation.

Gcoupler reveals endogenous, intracellular Ste2p allosteric modulators

We next utilized Gcoupler to test the hypothesis that the intracellular metabolites could potentially and directly regulate the GPCR signaling by directly interacting with the GPCR-Ga-protein interaction interface (Figure 1l). To test this hypothesis, we utilized a well-characterized yeast mating pathway mediated via the Ste2p-Gpa1p interface (Supplementary Figure 4a). Gcoupler was employed to screen metabolites from the Yeast Metabolome Database (YMDB) against the yeast phospholipid bilayer-embedded molecular dynamics simulated cryo-EM structure of Ste2p (Supplementary Figure 4b-i). This led to the identification of 17 potential surface cavities on Ste2p, with two intracellular regions, IC4 and IC5, accounting for over 95% of the Ste2p-Gpa1p interface (Figure 2a-b, Supplementary Figure 4j-n). Conservation analysis of these cavities across 14 yeast species further confirmed their structural significance (Supplementary Figure 4o). We independently ran Gcoupler for IC4 and IC5, compared their chemical diversity to ligands for the extracellular cavity (EC1), revealing distinct pharmacophore properties, and confirmed cavity-specific chemical spaces through principal component analysis (PCA) (Supplementary Figure 5a-e). We further assessed the reproducibility of the Gcoupler by

synthesizing 100 in silico compounds per run across five runs for the Ste2p IC4 cavity. Analysis of chemical diversity via 2D/3D PCA and pairwise Tanimoto Similarity (atom-pair fingerprints) revealed heterogeneous yet overlapping chemical compositions among ligands across runs (**Supplementary Figure 5f-h** [↗](#)).

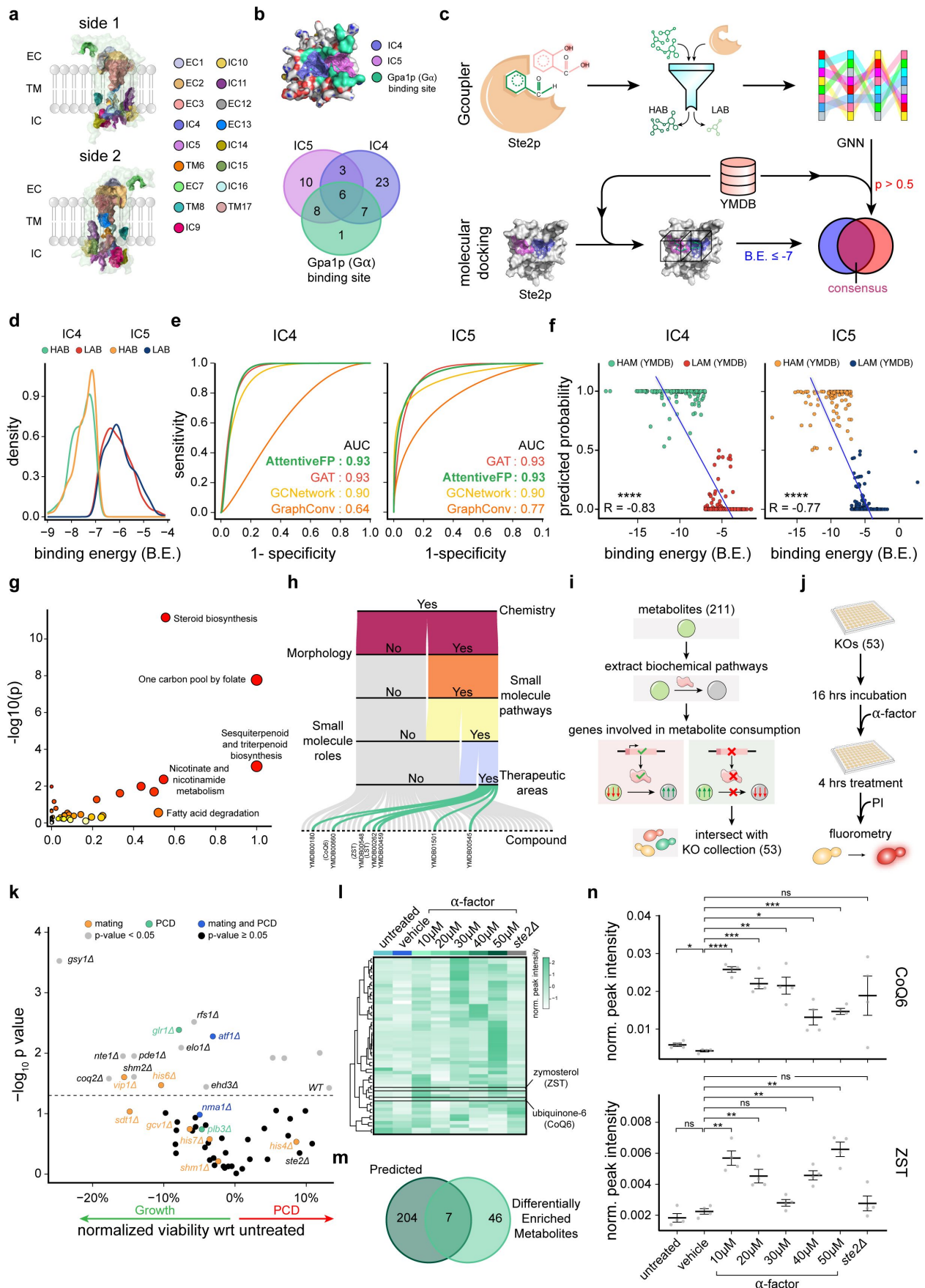


Figure 2

Identification of Endogenous, Intracellular Allosteric Modulators of Ste2p Using Gcoupler

(a) Schematic diagram depicting the topology of all the cavities identified using the Synthesizer module of the Gcoupler python package. Of note, the cavity nomenclature includes the cavity location, i.e., EC (extracellular), IC (intracellular), and TM (transmembrane), succeeded by numerical number. (b) Diagram depicting the three-dimensional view of the Ste2 protein, with highlighted Gα-protein binding site (Gpa1) and the Gcoupler intracellular cavities (IC4 and IC5). The venn diagram at the bottom depicts the percentage overlap at the amino acid levels between the Gα-binding site and predicted IC4 and IC5. (c) Schematic representation of the overall workflow used to predict the endogenous intracellular allosteric modulators of Ste2 receptor using Gcoupler and molecular docking technique. Of note, Yeast Metabolome DataBase (YMDB) metabolites were used as query compounds. (d) Overlapping density plots depicting and comparing the distributions of synthetic compounds predicted to target the IC4 and IC5 of the Ste2 receptor using the Gcoupler package. Of note, the Authenticator module of Gcoupler segregated the synthesized compound for each cavity (IC4 or IC5) into High-Affinity Binders (HAB) and Low-Affinity Binders (LAB). (e) AUC (Area under the curve) plots representing the performance of the indicated models. Notably, the models were trained using the cavity-specific synthetic compounds generated using the Gcoupler package. (f) Scatter plots depicting the relationship (correlation) between the binding prediction probabilities using Gcoupler and binding free energies computed using molecular docking (AutoDock). (g) Scatterplot depicting the Pathway Over Representation Analysis (ORA) results of the endogenous metabolites that were predicted to bind to the GPCR-Gα-protein (Ste2p-Gpa1p) interface using both Gcoupler and molecular docking. (h) Alluvial plot showing five level sub-activity spaces screening of the selected metabolites for IC4. (i) Schematic diagram depicting the workflow opted to narrow down on the single metabolic gene mutants. (j) Schematic diagram depicting the experimental design used to screen single metabolic gene mutants for α-factor-induced PCD. Cell viability was assessed using Propidium iodide-based fluorometric assay. (k) Scatter plot depicting the impact of α-factor stimuli on cellular viability, assessed using Propidium iodide-based fluorometric assay. The y-axis represents $-\log_{10}(\text{p-value})$ of the one-sample student's t-test between the normalized PI fluorescence of untreated and treated conditions. The x-axis represents the percentage inhibition or increase in cellular viability, estimated using a Propidium Iodide-based assay. The mutants reported to be involved in mating, PCD, or both are indicated in orange, green, and blue, respectively. The statistically non-significant mutants are indicated below the dashed line in black. (l) Heatmap depicting the relative enrichment/de-enrichment of differentially enriched metabolites in the indicated conditions. Of note, four biological replicates per condition were used in the untargeted metabolomics. (m) Venn diagram depicting the overlap between the predicted endogenous intracellular allosteric modulators of Ste2p and differentially enriched metabolites (DEMs) identified using untargeted metabolomics. (n) Mean-whisker plot depicting the relative abundance of ubiquinone 6 (CoQ6) and zymosterol (ZST) in the indicated conditions. Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance.

Post these performance/reproducibility checks of Gcoupler, we finally generated high-affinity (HABs) and low-affinity (LABs) binders, screened yeast metabolites using a hyperparameter-tuned Graph Neural Network (Attentive FP), and predicted potential binders to IC4 and IC5 with a >0.5 binding probability. AutoDock-based molecular docking of YMDB metabolites against the cavities further refined our list, revealing metabolites with consensus binding energies ≤ -7 kcal/mol and binding probabilities >0.5 (Figure 2c-f, Supplementary Figure 6a-f, Supplementary Table 4). Furthermore, HAMs (High-Affinity Metabolites) and LAMs (Low-Affinity Metabolites) displayed distinct atomic fingerprints, with enriched functional groups, including R2NH, R3N, ROPO3, ROH, and ROR, observed in HAMs (Supplementary Figure 6g-h). Pathway enrichment analysis identified metabolites involved in steroid, sesquiterpenoid, and triterpenoid biosynthesis, along with folate pathways (Figure 2g). BioRanker module further pinpointed sterols, including zymosterol (ZST), ubiquinone 6 (CoQ6), and lanosterol (LST), as top candidates, exhibiting high prediction probabilities (>0.99) and structural similarity to HABs (Figure 2h, Supplementary Figure 6i-j). To validate Gcoupler-identified allosteric modulators, we performed control

analyses of the Authenticator and Generator modules and blind docking of YMDB metabolites with Ste2p. Removing HAB and LAB labels from IC4 cavity compounds and applying random data splitting yielded poor model performance compared to Authenticator-guided splitting, confirming its robustness (**Supplementary Figure 6k** [↗](#)). Increasing training data size (25%, 50%, 75%, 100%) improved model accuracy for YMDB metabolite projections (**Supplementary Figure 6l-m** [↗](#)). Blind docking using AutoDock failed to distinguish HAM and LAM, unlike cavity-specific analysis via AutoDock and Gcoupler, which showed clear segregation (binding energy cutoff: -7 kcal/mol, Gcoupler probability: 0.5; **Supplementary Figure 6n** [↗](#)). All these rigorous control analyses and blind docking validated the reliability of Gcoupler's predictions, confirming its robustness in identifying cavity-specific modulators.

To experimentally validate the role of Gcoupler-predicted metabolites in Ste2p signaling, we performed a genetic screen of metabolic mutants. Mapping the predicted metabolites to KEGG and MetaCyc pathways revealed key enzymes involved in their metabolism. Screening 53 single metabolic deletion mutants showed that most deletion mutants resisted α -factor-induced PCD, with some displaying accelerated growth, suggesting a metabolic modulation of Ste2 signaling (**Figure 2i-k** [↗](#), **Supplementary Figure 7a-c**, **Supplementary Table 5** [↗](#)). Notably, the *STE2* knockout mutant (*ste2Δ*) displayed no significant PCD induction, validating the importance of Ste2p in this process. To further investigate the metabolic pathways associated with PCD resistance, we conducted high-resolution metabolomics on yeast cells exposed to varying α -factor concentrations. Survivors of PCD induction exhibited differential metabolite profiles, with CoQ6 and ZST showing consistent enrichment across conditions (**Figure 2l-n** [↗](#), **Supplementary Figure 7d-j**, **Supplementary Table 6** [↗](#)). Pathway analysis of these metabolites indicated involvement in glyoxylate, dicarboxylate, purine, and vitamin B6 metabolism (**Supplementary Figure 7k** [↗](#)). These findings, in conjunction with the genetic screen, highlight the interplay between metabolism and Ste2 signaling, with computationally predicted metabolites like ZST and CoQ6 potentially conferring resistance to α -factor-induced PCD.

Elevated Endogenous Metabolic levels selectively inhibit GPCR signaling

To evaluate the stability of the interactions of zymosterol (ZST), lanosterol (LST), and ubiquinone 6 (CoQ6) at the Ste2p-Gpa1p interface, we performed molecular dynamics (MD) simulations (100 ns) for Ste2p-metabolite complexes. Results revealed thermodynamically stable interactions across cavities IC4 and IC5, except for CoQ6, which exhibited fluctuating RMSD in IC5 (**Figure 3a-c** [↗](#), **Supplementary Figure 8a-b** [↗](#)). Extended 550 ns simulations confirmed stability for ZST and LST in IC4, while CoQ6 showed RMSD fluctuations post 100 ns (**Supplementary Figure 8c** [↗](#)). To gain further insight into the contributing residues from the MD simulations, we performed a residue-wise decomposition analysis that provides information about the energy contributions from the different residues to total binding free energies (**Supplementary Figure 9a** [↗](#)). These results suggest that IC4 and IC5 specific residues predominantly contribute to the total binding free energies. Notably, the binding free energies are obtained as an average of over 500 configurations corresponding to the last 50 ns of the MD simulations.

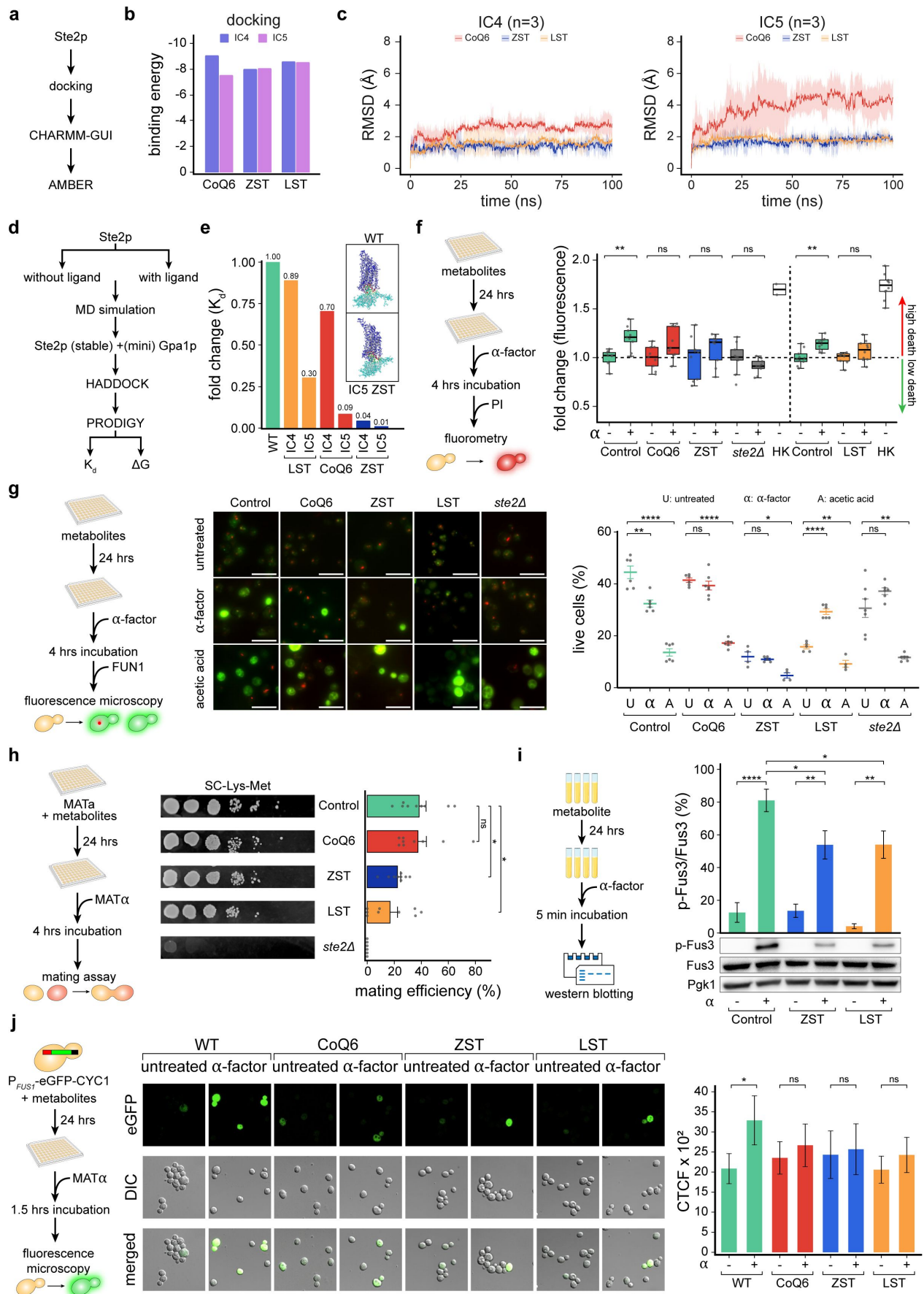


Figure 3

Elevated Endogenous Metabolite Levels Stabilize Ste2p-Gpa1p Interactions and Selectively Inhibit GPCR Signaling

(a) Scheme representing the key steps opted for preparing Ste2p structure for downstream computational analysis. (b) Barplots depicting the binding energies obtained by the docking of Ste2p and indicated metabolites across IC4 and IC5. (c) Line plots depicting the Root Mean Square Deviation (RMSD) changes over simulation timeframes from the three independent replicates of the indicated conditions in the indicated conditions. The spread of the data is indicated as Standard Deviation (SD). Notably, RMSD is provided in Angstroms (Å), whereas the simulation time is in nanoseconds (ns). (d) Workflow depicting the steps involved in Ste2p-miniG-protein docking using HADDOCK and PRODIGY web servers. (e) Barplots depicting the fold change of the dissociation constant (Kd) in the indicated conditions. Notably, fold change was computed with respect to the wild-type condition (Ste2p-miniG-protein). Insets represent molecular representations of Ste2p-miniG-protein and the highlighted interface residues. (f) The schematic diagram depicting the experimental workflow used to quantify α -factor-induced PCD using a propidium iodide-based cell viability fluorometric assay. Box plot on the right depicting the rescue from the α -factor-induced PCD in the indicated conditions as inferred using propidium iodide-based cell viability fluorometric assay (n=9 or 10 biological replicates; Heatkilled = 2). The y-axis represents the fold change of the propidium iodide fluorescence values with respect to their respective controls. Mann Whitney U test was used to calculate statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. (g) Schematic representation (left) of the experimental approach used to measure cell vitality and viability using microscopy-based FUNTM1 staining. Representative micrographs (right) depicting the FUNTM1 staining results in the indicated conditions, Scale 10 μ m. Mean-whisker plot depicting the relative proportion of the vital and viable yeast cells observed using FUNTM1 staining in the indicated conditions (n = 3 biological replicates). A Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. Error bars represent the standard error of the mean (SEM). (h) Schematic representation (left) of the experimental design for the mating assay (n = 3 biological replicates, each with three technical replicates). MAT α yeast cells were preloaded with the metabolites and then mated with MAT α cells to evaluate the mating efficiency. Representative micrographs in the middle qualitatively depict the mating efficiency in the indicated conditions. The bar plots on the right depict the mating efficiency (mean $\pm$ SEM) in the indicated conditions. Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. (i) Schematic representation depicting the experimental design of phospho-MAPK activity-based Western blot. Barplots depicting the p-Fus3 levels (mean $\pm$ SEM) in the indicated conditions. Error bars representing standard error of mean (SEM). A Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. (j) Schematic representation (left) of the experimental approach used to measure the fluorescence in *P_{FUS1}-eGFP-CYC1* yeast cells. Representative micrographs (right) depicting the eGFP expression in the yeast cells in the indicated conditions. Scale 20 μ m. Barplot depicting the Corrected Total Cell Fluorescence (CTCF) value (mean $\pm$ SEM) in the indicated conditions. A Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance.

To gain deeper insight into the mode of action of these metabolites in inhibiting Ste2p signaling, we first analyzed their impact at the orthosteric site, i.e., α -factor binding. We performed protein-peptide docking of Ste2p and α -factor and observed that metabolite-binding at the Ste2p-Gpa1p interface favors the α -factor interaction, as inferred from the binding free energies ΔG (kcal/mol) (Supplementary Figure 9b-c). We also analyzed protein-protein interaction between the Ste2p (GPCR) and miniGpa1p (55) by selecting GPCR configurations with and without metabolite-induced altered cavity topologies (IC4 and IC5), respectively (frames output from the aforementioned Ste2p-metabolite complex simulations) and computed the dissociation constant (Kd), binding affinity (ΔG) and the structural changes in the overall Ste2p topology (Figure 3d-e, Supplementary Figure 9b-d). These computational analyses revealed that in contrast to metabolite-free Ste2p-Gpa1p interaction, referred to as wild-type (WT) condition, the Kd value is multi-fold lower in the presence of metabolite, indicating a cohesive response induced by these

metabolites. A multi-fold lower K_d value further indicates and potentially explains that the metabolites binding favors Ste2p (GPCR) and miniGpa1-protein interaction and enables the establishment of a stable complex that might influence the shielding of the effector-regulating domains of the Gpa1p or influence its binding with the Ste4p (G β)-Ste18p (G γ) complex.

We next asked whether the observed resistance toward α -factor-induced PCD in single metabolic mutants might be the direct consequence of the identified metabolites or a pleiotropic response due to an altered genome. To test this, we next exogenously elevated ZST, CoQ6, and LST levels (prediction probability > 0.99 and post-filtration using BioRanker) in wild-type yeast and observed selective modulation of Ste2p signaling. Pre-loaded cells exhibited rescue from α -factor-induced PCD across growth kinetics, PI-based viability, and FUNTM1 assays while maintaining α -factor specificity over acetic acid-induced PCD (**Figure 3f-g** [↗](#), **Supplementary Figure 10a-c** [↗](#)). Mating assays further supported reduced Ste2p signaling in metabolite-preloaded cells, with CoQ6 showing no significant effects, consistent with its instability observed in molecular dynamics simulations (**Figure 3h** [↗](#), **Supplementary Figure 10d** [↗](#)). Further, we evaluated the deactivation of the pathway at the MAPK signaling level, where the α -factor-induced p-Fus3 levels were suppressed by ZST and LST (**Figure 3i** [↗](#)) but not CoQ6 (data not shown). Moreover, by using the P_{FUS1}-eGFP-CYC1 strain-based reporter assays, metabolite pre-treatment confirmed the inhibition of α -factor signaling (**Figure 3j** [↗](#)). These findings suggest endogenous metabolites as modulators of Ste2p signaling by stabilizing Ste2p-Gpa1p interactions.

Site-Directed Ste2 Mutants Abrogate Metabolite-Mediated Rescue Phenotype

To gain a deeper understanding, we investigated the role of metabolite binding in modulating Ste2p-Gpa1p interaction dynamics and employed both computational and experimental approaches. *In silico* site-directed mutagenesis of Ste2p revealed key metabolite-binding residues: S75 and R233 for CoQ6, L289 for ZST, and T155, V152, and I153 for LST. Prioritizing strong hydrogen bonds within an ideal 2.7–3.3 Å range, we selected mutants S75A, L289K, and T155D. These mutations significantly increased the dissociation constant (K_d) of the Ste2p-Gpa1p complex, indicating weakened interactions compared to wild-type Ste2p (**Figure 4a-b** [↗](#), **Supplementary Figure 11a** [↗](#)). This computational evidence supports the hypothesis that metabolite binding stabilizes the Ste2p-Gpa1p complex, facilitating the rescue response to α -factor-induced programmed cell death (PCD).

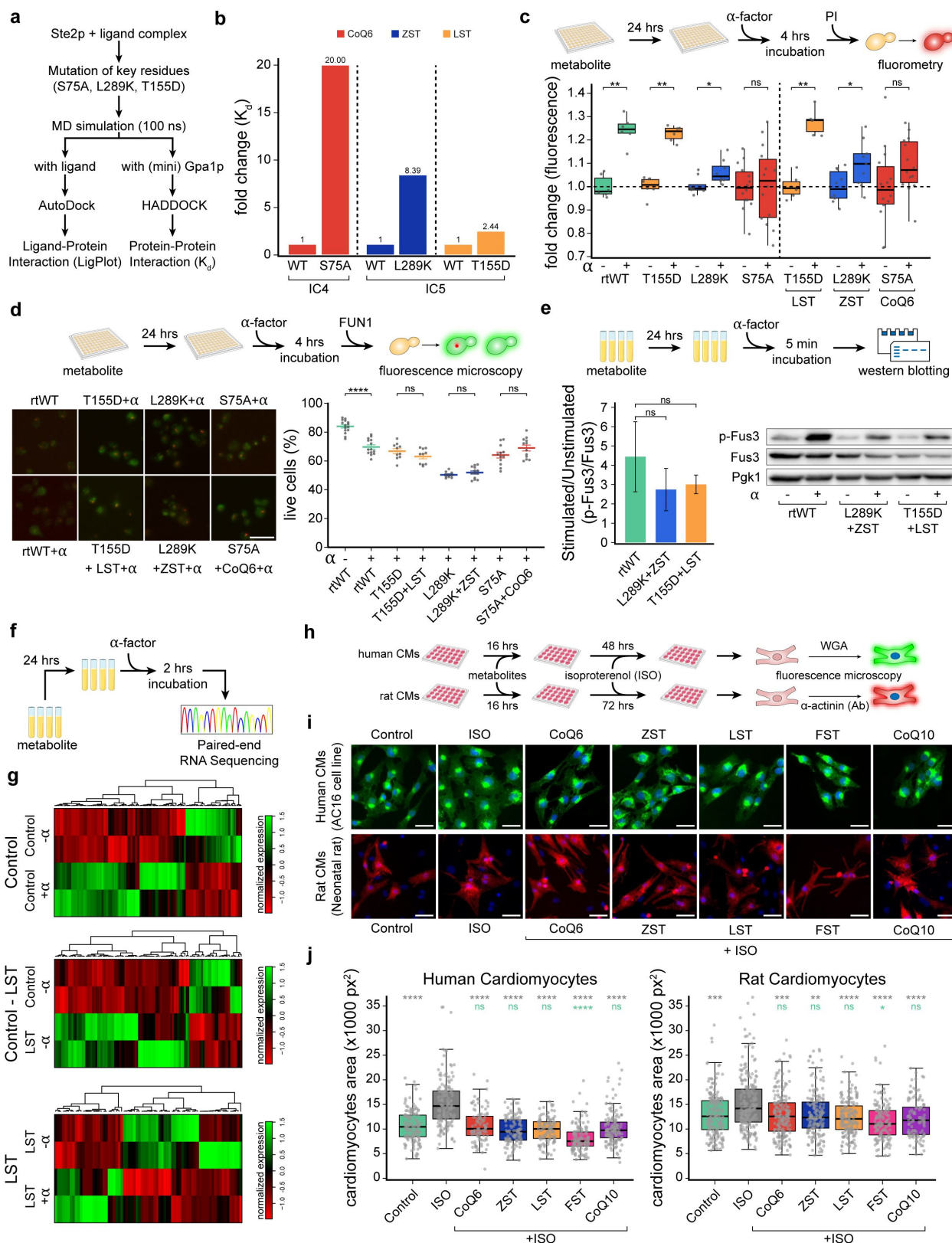


Figure 4

Site-Directed Ste2p Mutants Disrupt Metabolite-Mediated Rescue

(a) Workflow depicting the steps involved in Ste2p-miniG-protein docking of the wild-type and site-directed Ste2p mutants. Notably, docking was performed using HADDOCK and PRODIGY web servers. **(b)** Barplots depicting the dissociation constant (K_d) fold change in Ste2p site-directed mutants and wild-type. Notably, fold change was computed with respect to the metabolite influenced wild-type condition (Ste2p-miniG-protein). **(c)** The schematic diagram depicting the experimental workflow used to quantify α -factor-induced programmed cell death in generated site-directed missense mutants (T155D, L289K, S75A), alongside reconstituted wild-type STE2 (rtWT), using a propidium iodide-based cell viability fluorometric assay. The box plot (left) depicts the increase in the relative proportion of dead cells upon α -factor exposure. Box plot (right) depicting the loss of rescue phenotype from the α -factor-induced programmed cell death in the indicated conditions when pre-loaded with metabolites as inferred using propidium iodide-based cell viability fluorometric assay. The y-axis represents the fold change of the propidium iodide fluorescence values with respect to their respective controls. Mann Whitney U test was used to calculate statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. **(d)** Schematic representation (top) of the experimental approach used to measure cell vitality and viability using microscopy-based FUNTM1 staining. Representative micrographs (below) depicting the FUNTM1 staining results in the indicated conditions, Scale 10 μ m. Mean-whisker plot depicting the relative proportion of the vital and viable yeast cells observed using FUNTM1 staining in the indicated conditions (n = 4 biological replicates). A Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. Error bars represent the standard error of the mean (SEM). **(e)** Schematic representation (up) depicting the experimental design of phospho-MAPK activity-based Western blot. Barplots (down) depicting the p-Fus3 levels (mean $\pm$ SEM) in the indicated conditions. The y-axis represents the p-Fus3/Fus3 ratio for the stimulated condition normalized by its corresponding unstimulated sample. A Student's t-test was used to compute statistical significance. Asterisks indicate statistical significance, whereas ns represents non-significance. **(f)** Schematic representation depicting the experimental design of RNA sequencing, featuring treatment duration and the sequencing parameters. **(g)** Heatmap depicting the expression of differentially expressed genes obtained from RNA sequencing in the indicated conditions. Notably, Control and LST represent yeast cells unloaded and pre-loaded with lanosterol, respectively. α -factor is represented as α , where plus and minus signs represent its presence and absence, respectively. **(h)** Schematic representation of the experimental workflow followed to deduce the impact of indicated metabolites treatment on isoproterenol (ISO)-induced, GPCR-mediated hypertrophy response in human (AC16) and neonatal rat cardiomyocytes. Notably, in the case of AC16 cells, Wheat germ agglutinin (WGA) was used to stain the cardiomyocytes, whereas, for neonatal cardiomyocytes, alpha-sarcomeric actinin staining was used. **(i)** Micrographs depicting the human (above; green colored) and neonatal rat (below; red colored) cardiomyocytes in the indicated conditions. Scale 50 μ m. **(j)** Box plots depicting the surface area of human (AC16) and neonatal rat cardiomyocytes in the indicated conditions. Statistical significance of indicated metabolites with untreated control and isoproterenol-treated conditions are indicated in green and grey text, respectively. Mann Whitney U test with Bonferroni corrected p-values was used to compute statistical significance.

Next, we performed experimental validation by generating site-directed mutants and confirmed these computational predictions. Mutants (S75A, L289K, T155D) expressed in a *ste2 Δ* background (**Supplementary Figure 11b**) exhibited no rescue from α -factor-induced PCD upon pre-treatment with their respective metabolites, as shown by fluorometry-based cell death assays, FUNTM1 staining, and p-Fus3 signaling analysis by mapping MAPK pathway activation (**Figure 4c-e**). The lack of rescue highlights the direct role of metabolite binding at the Ste2p-Gpa1p interface in regulating downstream signaling. Interestingly, S75A mutants showed no α -factor-induced effects, likely due to significant structural disruptions. Shmoo formation assays further corroborated these findings, with no rescue effects observed in the mutants despite metabolite pre-loading (**Supplementary Figure 11c-e**).

To gain mechanistic insights, we performed RNA sequencing on LST-preloaded and untreated (control) cells with and without α -factor exposure. Transcriptomic analysis revealed differential expression of genes implicated in PCD and mating responses, including *GSY1*, whose downregulation was linked to α -factor resistance (**Figure 4f-g**, **Supplementary Figure 12a-c**, **Supplementary Tables 7-10**). Loss-of-function studies using knockout strains validated the involvement of key transcripts in metabolite-mediated rescue, particularly *YCR095W-A*, which exhibited a pronounced phenotype loss (**Supplementary Figure 12d-f**). These results collectively suggest that metabolite binding at the Ste2p-Gpa1p interface directly drives rescue responses, with secondary contributions from differentially expressed genes in attenuating α -factor-induced cell death.

Finally, we explored whether intracellular allosteric modulators such as ubiquinone 6 (CoQ6), zymosterol (ZST), lanosterol (LST), fucosterol (FST), and ubiquinone 10 (CoQ10) modulate GPCR signaling in higher vertebrates such as human and rat beta 1/2-adrenergic receptors signaling. Notably, computational docking at the GPCR-G α interface revealed strong binding affinities for these metabolites, similar to Ste2p-metabolite interactions (**Supplementary Figure 13a-b**). Sequence conservation analysis of the GPCR-G α interface across yeast Ste2p and adrenergic receptors in humans and rats further confirmed a high degree of evolutionary conservation at the metabolite-binding residues (**Supplementary Figure 13c**). Finally, to test this functional relevance, we evaluated the effect of these metabolites on isoproterenol-induced adrenergic receptor-mediated cardiac hypertrophy in human AC16 cardiomyocytes and neonatal rat cardiomyocytes. Preloading cells with these metabolites significantly attenuated hypertrophic responses, as evidenced by reduced single-cell surface area in quantitative assessments (**Figure 4h-j**). Notably, to further evaluate the evolutionary conservation of this phenomenon, we also analyzed 75 unique GPCR-G α complex structures from six species, selected from the PDB database. Dynamic docking was performed using five metabolites (CoQ6, ZST, LST, FST, and CoQ10) identified by Gcoupler as potential allosteric modulators and five negative controls predicted as poor binders. Results revealed significantly lower docking scores (< -7 kcal/mol) for Gcoupler-recommended metabolites compared to negative controls, irrespective of GPCR type or species (**Supplementary Figure 13d-h**). These findings demonstrate that intracellular metabolite modulation of GPCR activity is a conserved mechanism extending beyond yeast to higher vertebrates.

Discussion

Over the last few decades, extensive research has focused on identifying allosteric modulators of GPCRs due to their relevance in drug discovery (7, 20). Most known modulators are exogenous and target extracellular sites, while intracellular allosteric sites, identified recently through structural biology, offer novel avenues for regulation (3, 4, 6–8, 20). These sites, overlapping with G-protein and β -arrestin coupling regions, highlight the potential for intracellular allosteric modulation (1, 5, 8). Intracellular modulators, including chemically diverse agents like auto-antibodies and sodium ions, remain poorly understood, emphasizing the need for systematic exploration of these sites (4–6, 8, 9, 11–13, 37). However, the lack of data on intracellular modulators limits the feasibility of conventional computational approaches (14, 19, 38)(39–42).

To address this gap, we developed Gcoupler, a computational framework integrating *de novo* cavity identification, ligand synthesis, statistical analysis, graph neural networks, and Bioactivity-based ligand prioritization. Unlike existing tools, Gcoupler does not require cavity-specific experimentally validated compounds for model training. Gcoupler's precision in cavity mapping, flexibility for user-defined queries, and ability to screen large chemical libraries make it a versatile and efficient tool. Additionally, Gcoupler's generic design allows application beyond GPCRs, contrasting with existing platforms that often have limitations in modularity, precision, or

open-source availability. Noteworthy, in contrast to other known allosteric sites identification tools for GPCRs, such as Allosite ([43](#)), AllositePro ([44](#)), AlloReverse ([45](#)), that largely leverage the ML-based models or require orthosteric ligand-bound structure as input, the cavity detection feature of Gcoupler (LigBuilderV3), a critical step for the entire workflow, is not limited to only the allosteric sites; instead, it identifies all possible cavity-like regions on the protein surface, which then gets classified into druggable, undruggable, or amphibious based on their individual scoring and ligandability, thus making it unbiased and more specific towards query protein. Notably, the rationale for opting for LigBuilder V3 for cavity identification over similar tools such as Fpocket ([46](#)) is that the former uses a hydrogen atom probe, moving along the protein surface grid of 0.5 Å for cavity detection, being much more precise in detecting cavity boundaries, in both breadth and depth mapping; in contrast, the latter considers clusters of alpha spheres (Supplementary Table 11).

To date, only a few methods leverage generative AI models for cavity/pocket-based drug design. Gcoupler is an open-source, end-to-end platform integrating Ligand-Based Drug Design (LBDD) and Structure-Based Drug Design (SBDD) for drug design and large-scale screening. Unlike Pocket Crafter ([47](#)), which requires proprietary tools (e.g., MOE QuickPrep) and lacks predictive model-building modules, Gcoupler offers comprehensive functionality. Similarly, DeepLigBuilder ([23](#)) and Schrodinger's AutoDesigner are either closed source or limited in features compared to Gcoupler. Comparative analysis highlights Gcoupler's unique advantages in precision, flexibility, and functionality (Supplementary Table 11).

Using Gcoupler, we investigated the molecular basis of innate resistance to α -factor-induced programmed cell death (PCD) in yeast. Unlike humans, yeast possess only two GPCR systems, making their pheromone-sensing pathway ideal for focused study. Previous research predominantly identified downstream regulatory mechanisms ([29](#), [48](#), [49](#)); however, our findings suggest an upstream, receptor-level regulation via endogenous intracellular metabolites. Computational and experimental evidence pinpointed specific metabolites binding to the Ste2p-miniGpa1 interface, modulating signaling. Site-directed mutagenesis confirmed the functional relevance of these metabolite-interacting residues. Of note, previous mutagenesis experiments also revealed multiple critical amino acid residues that overlap with IC4 and IC5, suggesting their functional relevance in Ste2p downstream signaling (Supplementary Table 12). Despite these advances, certain limitations remain, including the potential pleiotropic effects in metabolic gene knockouts and challenges in replicating natural metabolite concentrations. Mechanistic insights into sterol biosynthesis mutants (*ergΔ*) revealed impaired mating responses due to heterogeneous defects, such as reduced sterol accumulation and shmoo formation, impaired membrane fusion, and decreased *FUS1* expression ([50](#)–[54](#)). This highlights a novel role for sterols in GPCR regulation and their broader implications for yeast microbial factories and stress tolerance ([55](#), [56](#)). Sterols and other endogenous metabolites were shown to modulate GPCR activity by targeting conserved G α -binding sites, reinforcing the evolutionary conservation of this mechanism across species, as demonstrated in human and rat hypertrophy models in this study.

In summary, our work uncovers a novel regulatory mechanism for GPCRs mediated by intracellular metabolites and presents a computational framework, Gcoupler, to explore unexplored allosteric sites. The proposed model suggests that selective metabolites binding to GPCR-G α interfaces induce local conformational changes, stabilizing GPCR-G-protein complexes and potentially obstructing downstream signaling. Alternative mechanisms, such as orthosteric site modulation, kinase/arrestin interaction interference, or alterations in membrane dynamics, remain to be explored, warranting further investigation. A critical limitation of our study is the absence of direct binding assays to validate the interaction between the metabolites and Ste2p. While our results from genetic interventions, molecular dynamics simulations, and docking studies strongly suggest that the metabolites interact with the Ste2p-Gpa1 interface, these findings remain indirect. Direct binding confirmation through techniques such as surface plasmon resonance, isothermal titration calorimetry, or co-crystallization would provide definitive

evidence of this interaction. Another critical limitation of our findings is its reliance on tools like AutoDock and PRODIGY for preliminary binding affinity estimates, which lack the thermodynamic precision of advanced methods. To address this, we employed MD simulations with MM/GBSA, incorporating factors like protein flexibility and solvation effects for more accurate ΔG calculations. While computationally intensive approaches were beyond this study's scope, we ensured reported ΔG values reflected system conformational flexibility by basing them on pre-simulated docked structures from molecular dynamics simulations. Further, our results suggest that the metabolite binds to the Ste2p-Gpa1 interface and modulates receptor activity upon pheromone stimulation, as supported by various assays. However, the precise sequence of interactions between Ste2p, the metabolite, and Gpa1 remains unexplored, as it requires sequential experiments beyond this study's scope. Taken together, addressing these limitations in future work would significantly strengthen our conclusions and provide deeper insights into the precise molecular mechanisms underlying the observed phenotypic effects.

Material and methods

Backend code for the Gcoupler

The back-end code for Gcoupler is implemented entirely in Python (3.8) and comprises four modules: Synthesizer, Authenticator, Generator, and BioRanker. Synthesizer employs LigBuilder V3.0 (32) for *de novo in silico* ligand synthesis, identifying protein cavities likely to be active or allosteric sites using a hybrid GROW-LINK approach with a Genetic Algorithm. The module autonomously selects one cavity for ligand synthesis based on user-defined residue positions. Using the CAVITY function of LigBuilder (27), it classifies 3D grid points around the protein into occupied, vacant, and surface points and integrates geometric and physicochemical properties to identify binding sites (57). Synthesizer outputs ligand structures in SMILES and PDBQT formats, alongside cavity grid coordinates for downstream modules. Authenticator validates the synthesized ligands using AutoDock Vina (1.2.3) for virtual screening (33). Binding energy calculations classify ligands into High-Affinity Binders (HABs) and Low-Affinity Binders (LABs), preserving balance for subsequent deep learning analysis. The default energy threshold is set to -7 kcal/mol, but users can explore alternative cutoffs, visualize distributions, and perform statistical comparisons within the workflow. These steps ensure precise identification and prioritization of ligand candidates for further analysis. The Authenticator uses the Kolmogorov-Smirnov test (58), Anderson-Darling test (59), and Epps-Singleton test (60) for hypothesis testing for the comparison of the distributions. The Authenticator module visualizes ligand distributions using overlapping density plots and Empirical Cumulative Distribution Function (ECDF) curves. If the default threshold fails to produce statistically meaningful separation, users can supply alternative negative datasets, such as decoys generated via Gcoupler's inbuilt RDKit Chem module or custom datasets (61). The Generator module builds deep learning-based classification models using the DeepChem (2.6.1) library (62). It accepts High-Affinity Binders (HABs) and Low-Affinity Binders (LABs)/decoys from the Authenticator module to train four graph-based models: Graph Convolution Model (GCM), Graph Convolution Network (GCN) (63), Attentive FP (AFP) (64), and Graph Attention Network (GAT) (65). Class imbalance is addressed through upsampling techniques. Generator tests all models using default hyperparameters, returning performance metrics for user selection. Hyperparameters can be tuned via manual settings or using default values followed by k-fold cross-validation. The final optimized model, trained on the complete synthetic dataset (HAB + LAB/decoys), enables large-scale screening of user-supplied compounds based on their SMILES representations. The BioRanker module performs post-prediction analysis for functional activity-based compound screening. Positively predicted compounds are selected using a stringent probability threshold or adaptive methods such as G-means and Youden's J statistic, which optimize sensitivity and specificity. The selected compounds are projected into biological activity spaces (Chemistry, Targets, Networks, Cells, Clinics) by comparing their biological activity descriptor vectors with those of HABs using cosine similarity (34). A modified

PageRank algorithm ranks compounds based on activity-specific scores, with support for multi-activity ranking to refine results based on user-defined biological properties, ensuring precise and context-relevant compound prioritization.

Additional information about the backend code for Gcoupler along with methodology for Runtime analysis, Sequence-structural-functional level analysis, Molecular Dynamics Simulation, Molecular docking (AutoDock), Functional enrichment analysis, and Protein-protein docking can be accessed in the [Supplementary Information](#).

Gcoupler Benchmarking

To assess batch effects across Gcoupler runs for a specific cavity, we utilized the standard Gcoupler Docker image. Intracellular cavity 4 (IC4) of the Ste2 protein of yeast was used for benchmarking. A total of 100 molecules were *in silico* synthesized by the Synthesizer module of Gcoupler iteratively. Post-generation, atom pair fingerprints (ChemmineR; R package) were calculated for the synthesized molecules from each run, and the data was visualized using Principal Component Analysis and pairwise comparison using Tanimoto Similarity (ChemmineR, R package).

For model benchmarking, Gcoupler was validated on GPCRs from the DUD-E dataset, alongside the information about the active ligands and their randomly selected number-matched decoys (35). Additionally, Gcoupler's performance in identifying experimentally elucidated allosteric sites and modulators was tested using PDB complexes obtained from the RCSB PDB database.

Metabolomics

Wildtype (BY4741) and *ste2Δ* yeast strains were grown in YPD medium at 30°C, 150 rpm, through primary and secondary cultures (16 hours each). Equal cell numbers (1.5 mL) were aliquoted into a 96-well deep well plate. α -factor (Sigma-Aldrich) was added at final concentrations of 10, 20, 30, 40, and 50 μ M (eight replicates each). DMSO served as the solvent control, while untreated WT and *ste2Δ* conditions received no treatment. Plates were incubated (30°C, 150 rpm, 4 hours) under a breathable membrane. A 50 μ L aliquot was taken for Propidium Iodide (PI; 11195, SRL) assay as described in [Supplementary Information](#). Following the PI assay, four pooled replicates were pelleted (6000 rpm, 5 min, RT), treated with zymolyase (40 U/mL, 1X PBS, 30°C, 1 hour), washed with PBS, and metabolomics analysis was performed. Data analysis included peak normalization, omission of metabolites with constant or >50% missing values, and kNN-based imputation (MetaboAnalyst). Data were IQR-filtered, and differentially enriched metabolites (DEMs) were identified by calculating log₂ fold change ($|\log_2\text{FC}| \geq 1$, $p < 0.05$ via Student's t-test). Pathway Over-Representation Analysis (ORA) was performed using MetaboAnalyst with hypergeometric or Fisher's exact tests to assess pathway enrichment against background metabolite distributions. Further details about methodology is available in [Supplementary Information](#).

Genetic Screening

Fifty-three knockout strains from the Yeast Deletion Collection, along with WT and *ste2Δ*, were treated with α -factor (30 μ M), while DMSO served as the solvent control. Plates were incubated for 4 hours. A 50 μ L aliquot was used to measure Propidium iodide-based cell viability assessment assay as described in [Supplementary Information](#). Fluorescence data were normalized to blank-adjusted OD₆₀₀, followed by two additional rounds of normalization with unstained and HK controls. Percentage fold change for the treated group was calculated relative to the untreated group, and statistical significance was determined using a one-sample Student's t-test. Further details about the methodology used is available in [Supplementary Information](#).

Pre-loading of Yeast cells with metabolite

Yeast cells were cultured in YPD medium at 30°C, 200 rpm for 16 hours in primary and secondary cultures. Equal cell densities (5 μ L) from secondary cultures were inoculated into 96-well plates containing 145 μ L YPD with metabolites Coenzyme Q6 (CoQ6, 9001500, Avanti® Polar Lipids), Zymosterol (ZST, 700068P, Avanti® Polar Lipids), and Lanosterol (LST, L5768, Sigma-Aldrich) at 0.1 μ M, 1 μ M, and 10 μ M concentrations. Plates were incubated for 24 hours at 30°C, 200 rpm, with multiple biological replicates. Ethanol-treated wells served as solvent controls. For site-directed *STE2* mutants, the mutants were grown in YPD for primary and secondary cultures, but the metabolite pre-loading was performed in YPGR instead of YPD to induce Ste2 expression. After pre-loading, the following assays were performed: growth kinetics, propidium iodide-based assay, FUNTM₁ staining, Mating assay, Phospho-MAPK activity based-western blot and Transgenic reporter assay. The detailed protocol for each of these assays is available in [Supplementary Information](#).

RNA-Sequencing

Yeast cells (BY4741) were cultured in YPD medium at 30 °C, 200 rpm, with or without lanosterol (LST, 1 μ M) in biological duplicates. Cells were subsequently treated or untreated with α -factor (30 μ M) for 2 hours. RNA was isolated following Mittal et al. (66). Sequencing quality was assessed using MultiFastQ, and paired-end reads were trimmed and aligned to the *S. cerevisiae* reference genome (ENSEMBL R64-1-1; GCA_000146045.2) using the Rsubread package (v2.6.4). Gene expression counts were generated via featureCounts and normalized using TMM (Supplementary Table 7). Differential expression analysis was performed with NOISeq (v2.38.0) (Supplementary Tables 8–10). Functional enrichment was assessed using the Gene Ontology Term Finder (v0.86) from the *Saccharomyces* Genome Database. Raw FASTQ files and normalized expression data are available on Zenodo. Additional details are available in the [Supplementary Information](#).

Site-Directed Mutagenesis

The gene encoding wild-type *STE2* was PCR amplified from the genome of *Saccharomyces cerevisiae* and further cloned into plasmid pRS304 under galactose-inducible *GAL1* promoter to generate plasmid pRS304-P_{GAL1}-STE2-CYC1 using Gibson assembly (67, 68). The mutants were generated by PCR amplifying the gene with primers consisting of respective mutations and cloned into plasmid pRS304 under *GAL1* promoter to generate pRS304-P_{GAL1}-STE2 S75A-CYC1, pRS304-P_{GAL1}-STE2 T155D-CYC1, and pRS304-P_{GAL1}-STE2 L289K-CYC1. Wild-type (rtWT) and mutant *STE2* were integrated by digesting the pRS304 vector with restriction enzyme BstXI to generate linearized plasmid and transformed into *Saccharomyces cerevisiae* BY4741 *ste2* Δ strain. Additional details about the methodology is available in [Supplementary Information](#).

Cardiomyocytes Hypertrophy Models

Human AC16 cardiomyocytes were cultured in DMEM-F12 (Thermo Scientific) with 12.5% fetal bovine serum (FBS) at 37°C and 5% CO₂. Cells were seeded in a 24-well plate for size measurements, treated after 24 hours with metabolites (CoQ6, ZST, LST, FST, CoQ10) at 2.5 μ M, and incubated overnight with 1% FBS. The medium was refreshed with fresh metabolites and isoproterenol (25 μ M) for 48 hours. Cells were washed with PBS, fixed with 4% paraformaldehyde, and stained with wheat germ agglutinin (Thermo Scientific) and DAPI. Images were captured using a Leica DMI 6000 B microscope at 20X magnification, and cell area was measured using ImageJ. Neonatal rat cardiomyocytes were isolated from 1-3-day-old SD rat pups using Collagenase Type II. After heart explantation and digestion, the cells were centrifuged and pre-plated for 90 minutes to remove fibroblasts. The cardiomyocytes were seeded in a gelatin-coated 24-well plate, incubated overnight with 2.5 μ M metabolites and 1% FBS, and then treated with metabolites (2.5 μ M) and isoproterenol (10 μ M) for 72 hours. Cells were fixed, stained with alpha-sarcomeric

actinin and DAPI, and images were captured using a Leica DMI 6000 B at 20X magnification. Cell area was quantified using ImageJ. Additional details about methodology is available in [Supplementary Information](#).

Statistical Analysis

Statistical analyses were performed using Past4 software or R-Programming. The Mann-Whitney U test was applied to compare medians between two distributions (non-parametric), while Student's t-test was used for pairwise comparisons of means. P-value correction was performed using the Bonferroni method when necessary. A significance threshold of 0.05 was set, with *, **, ***, and **** indicating p-values <0.05, <0.01, <0.001, and <0.0001, respectively.

Data Availability

The processed untargeted metabolomics data is provided as Supplementary Information. The raw RNA sequencing files are available at ArrayExpress under accession *E-MTAB-12992*.

Code Availability

A Python package for Gcoupler is provided via pip <https://test.pypi.org/project/Gcoupler/>. A docker container pre-compiled with Gcoupler and all of its dependencies can be found at <https://hub.docker.com/r/sanjayk741/gcoupler>. The source code of Gcoupler is available on the project GitHub page: <https://github.com/the-ahuja-lab/Gcoupler> and also at Zenodo with DOI: 10.5281/zenodo.7835335, whereas the raw sequencing files can be accessed using DOI: 10.5281/zenodo.7834294.

Ethical statement

The local IAEC (Institutional Animal Ethics Committee) committee at CSIR-Central Drug Research Institute approved all the animal experiments (IAEC/2020/38) following the guidelines of the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), New Delhi, Government of India.

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Additional information

Author Contributions

The study was conceived by G.A. Computational analysis workflows were designed by G.A. Statistical guidance was provided by D.S. Docking, and Molecular Simulations were supervised by N.A.M and conducted by N.A.M, and S.K.M. Data analyses was performed by S.K.M., A.M., S.S., S.K., S.D., A.K.S., S.A., V.G., T.S.G., N.K.D and K.S.. S.K.M built the Gcoupler Python package. G.A. and A.M. designed yeast experimental workflows and performed by A.M. Site-directed yeast mutants were created by N. and D.S. Human cell culture-based experiments were performed and analyzed by A.G., and S.K.G. Illustrations were drafted by A.M., S.K.M., and G.A. G.A. wrote the paper. All authors have read and approved the manuscript.

Additional files

Supplementary Information. *The supplementary Information provides details of the nucleotide and amino acid Multiple Sequence Alignments (MSAs) of the wild-type STE2 and the site-directed missense mutants of STE2. Each row starts with the sequence identifier followed by the aligned sequence (in chunks) with the ending position of the aligned sequence provided at the end, separated by tabs. The mutation sites are highlighted in red (S75A), yellow (T155D) and blue (L289K). Supplementary Information also contains detailed materials and methods.* [↗](#)

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Reviewer #1 (Public review):**Summary**

In this manuscript, the authors introduce Gcoupler, a Python-based computational pipeline designed to identify endogenous intracellular metabolites that function as allosteric modulators at the G protein-coupled receptor (GPCR) - Ga protein interface. Gcoupler is comprised of four modules:

- I. Synthesizer - identifies protein cavities and generates synthetic ligands using LigBuilder3
- II. Authenticator - classifies ligands into high-affinity binders (HABs) and low-affinity binders (LABs) based on AutoDock Vina binding energies
- III. Generator - trains graph neural network (GNN) models (GCM, GCN, AFP, GAT) to predict binding affinity using synthetic ligands
- IV. BioRanker - prioritizes ligands based on statistical and bioactivity data

The authors apply Gcoupler to study the Ste2p-Gpa1p interface in yeast, identifying sterols such as zymosterol (ZST) and lanosterol (LST) as modulators of GPCR signaling. Our review will focus on the computational aspects of the work. Overall, we found the Gcoupler approach interesting and potentially valuable, but we have several concerns with the methods and validation that need to be addressed prior to publication/dissemination.

(1) The exact algorithmic advancement of the Synthesizer beyond being some type of application wrapper around LigBuilder is unclear. Is the grow-link approach mentioned in the methods already a component of LigBuilder, or is it custom? If it is custom, what does it do? Is the API for custom optimization routines new with the Synthesizer, or is this a component of LigBuilder? Is the genetic algorithm novel or already an existing software

implementation? Is the cavity detection tool a component of LigBuilder or novel in some way? Is the fragment library utilized in the Synthesizer the default fragment library in LigBuilder, or has it been customized? Are there rules that dictate how molecule growth can occur? The scientific contribution of the Synthesizer is unclear. If there has not been any new methodological development, then it may be more appropriate to just refer to this part of the algorithm as an application layer for LigBuilder.

(2) The use of AutoDock Vina binding energy scores to classify ligands into HABs and LABs is problematic. AutoDock Vina's energy function is primarily tuned for pose prediction and displays highly system-dependent affinity ranking capabilities. Moreover, the HAB/LAB thresholds of -7 kcal/mol or -8 kcal/mol lack justification. Were these arbitrarily selected cutoffs, or was benchmarking performed to identify appropriate cutoffs? It seems like these thresholds should be determined by calibrating the docking scores with experimental binding data (e.g., known binders with measured affinities) or through re-scoring molecules with a rigorous alchemical free energy approach.

(3) Neither the Results nor Methods sections provide information on how the GNNs were trained in this study. Details such as node features, edge attributes, standardization, pooling, activation functions, layers, dropout, etc., should all be described in detail. The training protocol should also be described, including loss functions, independent monitoring and early stopping criteria, learning rate adjustments, etc.

(4) GNN model training seems to occur on at most 500 molecules per training run? This is unclear from the manuscript. That is a very small number of training samples if true. Please clarify. How was upsampling performed? What were the HAB/LAB class distributions? In addition, it seems as though only synthetically generated molecules are used for training, and the task is to discriminate synthetic molecules based on their docking scores. Synthetic ligands generated by LigBuilder may occupy distinct chemical space, making classification trivial, particularly in the setting of a random split k-folds validation approach. In the absence of a leave-class-out validation, it is unclear if the model learns generalizable features or exploits clear chemical differences. Historically, it was inappropriate to evaluate ligand-based QSAR models on synthetic decoys such as the DUD-E sets - synthetic ligands can be much more easily distinguished by heavily parameterized ligand-based machine learning models than by physically constrained single-point docking score functions.

(5) Training QSAR models on docking scores to accelerate virtual screening is not in itself novel (see here for a nice recent example: <https://www.nature.com/articles/s43588-025-00777-x>), but can be highly useful to focus structure-based analysis on the most promising areas of ligand chemical space; however, we are perplexed by the motivation here. If only a few hundred or a few thousand molecules are being sampled, why not just use AutoDock Vina? The models are trained to try to discriminate molecules by AutoDock Vina score rather than experimental affinity, so it seems like we would ideally just run Vina? Perhaps we are misunderstanding the scale of the screening that was done here. Please clarify the manuscript methods to help justify the approach.

(6) The brevity of the MD simulations raises some concerns that the results may be over-interpreted. RMSD plots do not reliably compare the affinity behavior in this context because of the short timescales coupled with the dramatic topological differences between the ligands being compared; CoQ6 is long and highly flexible compared to ZST and LST. Convergence metrics, such as block averaging and time-dependent MM/GBSA energies, should be included over much longer timescales. For CoQ6, the authors may need to run multiple simulations of several microseconds, identify the longest-lived metastable states of CoQ6, and perform MM/GBSA energies for each state weighted by each state's probability.

<https://doi.org/10.7554/eLife.106397.1.sa2>

Reviewer #2 (Public review):

Summary:

Mohanty et al. present a new deep learning method to identify intracellular allosteric modulators of GPCRs. This is an interesting field for e.g. the design of novel small molecule inhibitors of GPCR signalling. A key limitation, as mentioned by the authors, is the limited availability of data. The method presented, Gcoupler, aims to overcome these limitations, as shown by experimental validation of sterols in the inhibition of Ste2p, which has been shown to be relevant molecules in human and rat cardiac hypertrophy models. They have made their code available for download and installation, which can easily be followed to set up software on a local machine.

Strengths:

- Clear GitHub repository
- Extensive data on yeast systems

Weaknesses:

- No assay to directly determine the affinity of the compounds to the protein of interest.

In conclusion, the authors present an interesting new method to identify allosteric inhibitors of GPCRs, which can easily be employed by research labs. Whilst their efforts to characterize the compounds in yeast cells, in order to confirm their findings, it would be beneficial if the authors show their compounds are active in a simple binding assay.

<https://doi.org/10.7554/eLife.106397.1.sa1>

Reviewer #3 (Public review):

Summary:

In this paper, the authors introduce the Gcoupler software, an open-source deep learning-based platform for structure-guided discovery of ligands targeting GPCR interfaces. Overall, this manuscript represents a field-advancing contribution at the intersection of AI-based ligand discovery and GPCR signaling regulation.

Strengths:

The paper presents a comprehensive and well-structured workflow combining cavity identification, de novo ligand generation, statistical validation, and graph neural network-based classification. Notably, the authors use Gcoupler to identify endogenous intracellular sterols as allosteric modulators of the GPCR-Gα interface in yeast, with experimental validations extending to mammalian systems. The ability to systematically explore intracellular metabolite modulation of GPCR signaling represents a novel and impactful contribution. This study significantly advances the field of GPCR biology and computational ligand discovery.

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